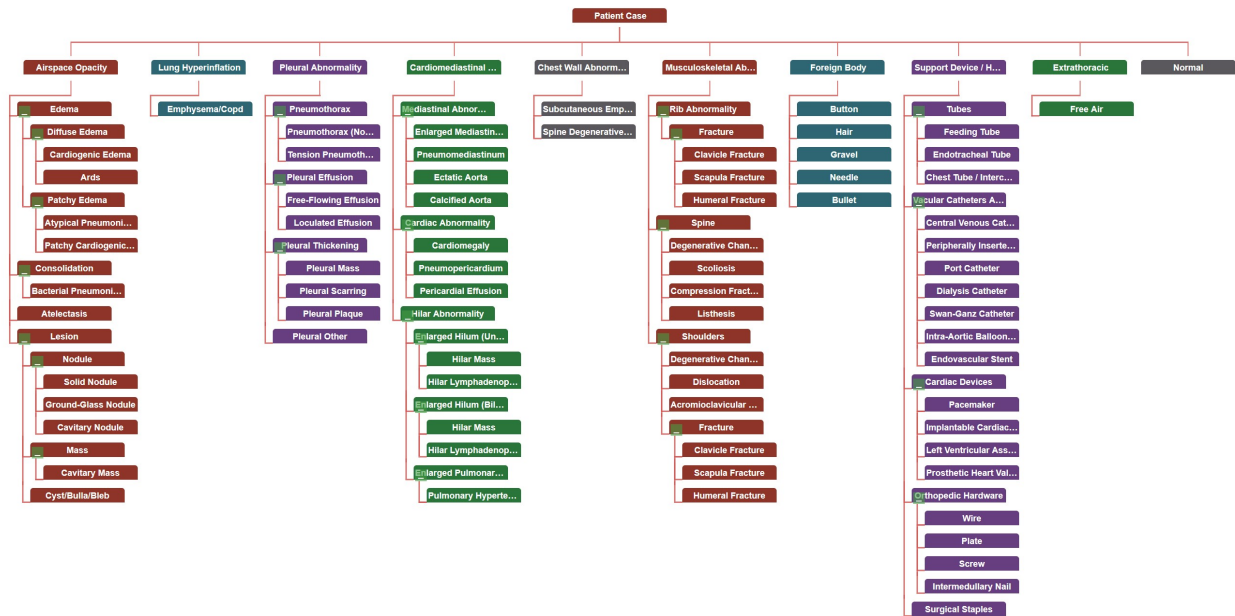


[The probability for the sub-pathologies is required only if the parent pathology prevalence is greater than 10%]

Figure B.4: Pathology Hierarchy



AI Support Tool

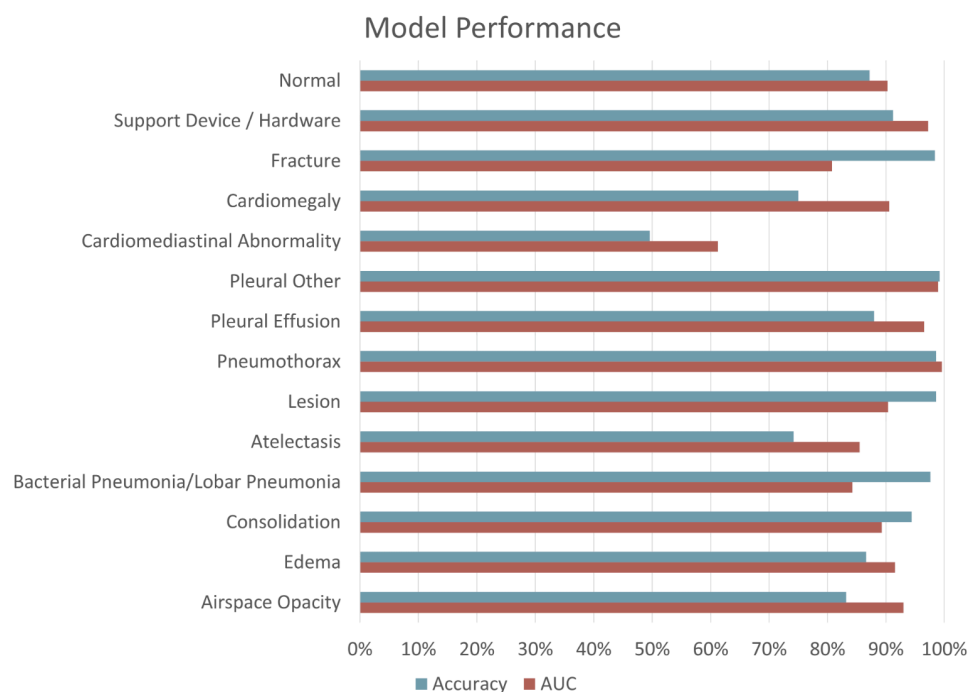
The AI support tool that is provided uses only the X-ray image to predict the probability of each potential pathology of interest. The tool is based on state-of-the-art machine learning algorithms developed by a leading team of researchers at Stanford University. The tool is trained only on X-ray images, meaning it does not incorporate the clinical history of the patients.

Performance of the AI Support

The AI tool is described in Irvin et al. [2019], which showed the AI tool performed at or near expert levels across the pathologies studied. Below we plot two measures of performance of the AI tool. We plot in blue the accuracy of the tool, defined as the share of cases correctly diagnosed when treating false positives and false negatives equally. In red, we plot the Area Under the ROC curve (AUC), which is another measure of AI classification performance.

The AUC is a number between 0 and 100%, with numbers close to 100% representing better algorithm performance. The AUC is equal to the probability that a randomly chosen positive case is ranked higher than a randomly chosen negative case.

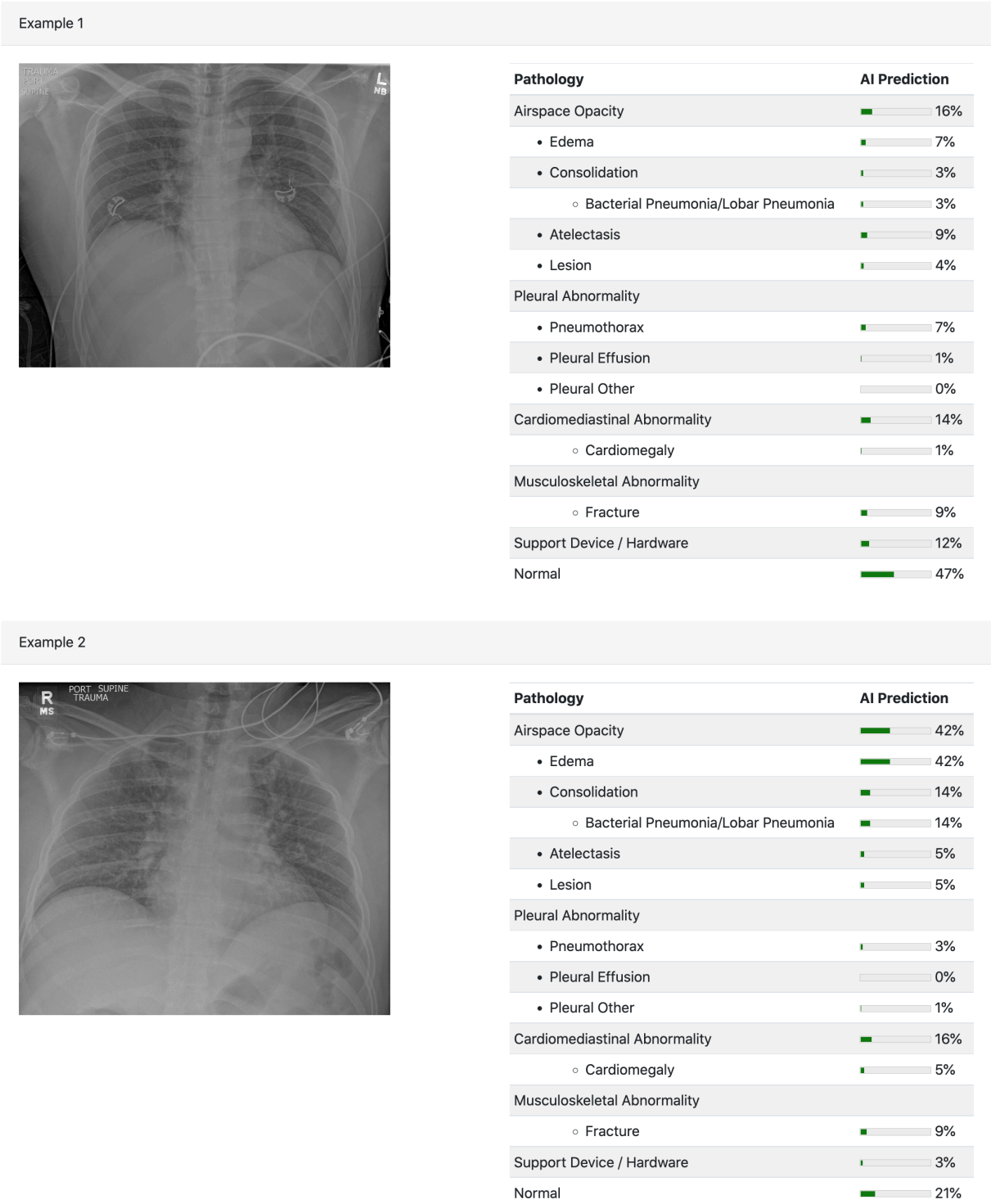
Figure B.5: Performance of AI Tool



Example Images

Below are 50 example images with the associated AI tool predictions. These images are randomly chosen to allow you to familiarize yourself with the AI support tool and its accuracy *[Here we only provide two out of the fifty images]*.

Figure B.6: Example Images



Demonstration

The brief video below walks you through the interface and a few examples. *[At this stage participants saw an instruction video which can be found [here](#)]*

Consent

You have been asked to participate in a study conducted by researchers from the Massachusetts Institute of Technology (M.I.T.) and Harvard University.

The information below provides a summary of the research. Your participation in this research is voluntary and you can withdraw at any time.

1. Study procedure: We will ask you to examine a number of chest x-rays. We will vary both the amount of information provided about the patient and the availability of an AI support tool.
2. Potential Risks & Benefits: There are no foreseeable risks associated with this study and you will receive no direct benefit from participating.

Your participation in this study is completely voluntary and you are free to choose whether to be in it or not. If you choose to be in this study, you may subsequently withdraw from it at any time without penalty or consequences of any kind. The investigator may withdraw you from this research if circumstances arise.

Privacy & Confidentiality

The only people who will know that you are a research subject are members of the research team which might include outside collaborators not affiliated with MIT. No identifiable information about you, or provided by you during the research, will be disclosed to others without your written permission, except: if necessary to protect your rights or welfare, or if required by law. In addition, your information may be reviewed by authorized MIT representatives to ensure compliance with MIT policies and procedures.

When the results of the research are published or discussed in conferences, no information will be included that would reveal your identity.

Questions

If you have any questions or concerns about the research, please feel free to contact us directly at diagnosticAI@mit.edu.

Your Rights

You are not waiving any legal claims, rights or remedies because of your participation in this research study. If you feel you have been treated unfairly, or you have questions regarding